## Systematic analysis of a simple salt

### Analysis of anions

**Salt NO:**

**Date:**

| Serial NO: | Experiment | Observation | Inference |
|---|---|---|---|
| 1 | Note the colour of the salt | Blue (or) Green<br>Brown | May be copper salt<br>May be an iron salt |
| 2 | **Action of heat:**<br>A small amount of a salt is strongly heated in a dry test tube. | A colourless gas with the pungent smell, forming a dense white fumes when a glass rod dipped in Conc. HCl is brought close to its mouth | Presence of an ammonium salt |
| | | A reddish brown gas is evolved | Presence of a nitrate salt |
| | | Salt is yellow when hot and white when cold | May be a zinc salt |
| 3. | **Flame test:**<br>Take a small amount of salt in a watch glass. Add a drop of Conc. HCl to it and form a paste. Take the paste at the charred end of the splinter and introduce it near the Bunsen flame | Bluish green flame | Presence of a copper salt |
| | | Apple green | Presence of a barium salt |
| | | Brick red | Presence of a calcium salt |
| 4. | **Action of dil. HCl:**<br>Take a small amount of salt in a test tube and add about 1mL of dil. HCl to it. Gently heat it in the Bunsen flame | Brisk effervescence is noted. A colourless, odourless gas turning lime water milky evolves | Presence of carbonate |
| | | A colourless gas with a rotten egg smell turning a paper dipped in lead acetate shining black evolves | Presence of sulphide |



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